Content, Not Vocabulary: Evaluating a Style Skill for LLM Prose Against Humanizers, Detectors, and Human Essays

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Abstract

Advice for making language-model prose read as human usually takes the form of banned-word lists, and "humanizer" tools rewrite text until detectors pass it. We test a prompt-level style skill that audits content and structure before vocabulary and forbids adding facts the writer did not supply. On 16 held-out briefs in ten genres, model judges ranked its writing above plain generation in 47 of 48 judgments, and above a word-list prompt and a popular humanizer in most, with about a third as many unsupported claims. Against 40 essays that admitted students wrote before ChatGPT, judges reading one essay at a time judged every real essay real, and the skill's essays read only slightly more human than the word-list prompt's: its rules had amplified a casual, hedged register. A version rebuilt on 24 of these essays and frozen before testing raised the share of ratings calling the other 16 real from 46% to 65% (the two stricter judges: 22% to 47%) and cut unsupported claims from 0.94 to 0.19 per essay, at a cost of 0.3 quality points; real essays stayed separable. Fixing the real essays' errors barely changed how real they looked. We release the skill, its versions and the evaluation code.

1 Introduction

Experienced readers have learned to recognize machine-written prose. Corpus studies trace sharp rises after the release of ChatGPT in words such as *delve* and *underscore* in biomedical abstracts (Kobak et al., 2025) and *commendable* and *meticulous* in peer reviews (Liang et al., 2024), and instruction-tuned models use present participial clauses at two to five times the human rate (Reinhart et al., 2025). Five people who use these tools heavily, voting together, misclassified only 1 of 300 articles, and they cited vocabulary, sentence structure and originality as their evidence (Russell et al., 2025).

Two responses dominate practice. The first is the banned-word list, distributed in prompts, style guides and community "skills". The second is the commercial humanizer, which rewrites AI-generated text to avoid automated or human detection (Roe et al., 2026). Both treat the surface as the problem. Wikipedia's editors, whose guide the most-starred humanizer skill is built on (Chen, 2026), warn against exactly this: "Please do not merely treat these signs as the problems to be fixed; that could just make detection harder" (Wikipedia contributors, 2026). Both responses also ignore a second failure of model prose: models fill gaps with plausible details that nobody supplied (Ji et al., 2023), and in personal writing those details are feelings and memories the writer never reported.

We built and evaluated an alternative: a style skill (a set of instructions an agent loads when a task matches) that orders its audit from content to staging, inflation, rhythm, tone and only then vocabulary, and that never adds a fact absent from the writer's material. For college application essays it switches to a coaching mode and does not write the essay.

Our contributions are:

- An evidence-graded design for a writing skill, built from a review of about 40 sources on AI-writing tells and of 16 existing open-source writing-skill repositories (11 of them humanizer or anti-slop skills), and released with versions 1 to 8 and a rejected candidate.
- A controlled comparison on 20 development and 16 held-out briefs in ten genres against plain generation, a word-list prompt and a popular humanizer, scored by judge panels under neutral labels (including a judge on a second model) and by three open-source detectors.
- A human-anchored benchmark: 40 admitted college essays written before ChatGPT,

model versions written from notes reverse-engineered from each, judges who read one essay at a time, a copyedit control, and a version frozen before a held-out test. On it, the version judges preferred among model texts read only slightly more human than a word-list prompt.

- A detector audit against human writing collected before ChatGPT. At least one detector scored 38 of 40 admitted essays at 20% or more, while all 89 model essays from invented briefs passed (and 4 of 40 plain essays written from real students' notes).
- A record of the iterations, including four versions or candidates that did not beat their predecessor, and a measurement effect: personal-statement rankings depend on which other texts share the panel.
- A coaching experiment comparing the skill's Coach mode with plain feedback on simulated student drafts.

2 Related work

Stylistic signatures of LLM text. Excess-vocabulary analyses of PubMed and arXiv abstracts and of peer reviews show that LLM adoption shifted the frequency of style words (Kobak et al., 2025; Liang et al., 2024; Juzek and Ward, 2025; Geng and Trotta, 2024), which in PubMed were mostly verbs and adjectives (Kobak et al., 2025). Beyond words, instruction-tuned models write a noun-heavy, informationally dense register (Reinhart et al., 2025), reuse syntactic templates from pre-training data (Shaib et al., 2024), and, in Mandarin, overuse the corrective "not X, but Y" construction, where over half of annotated uses (52.8%) were correct but unneeded (Peng et al., 2026). Professional writers editing model-written creative paragraphs agree on a seven-category taxonomy of idiosyncrasies, led by awkward word choice (28%) and poor sentence structure (20%) ahead of unnecessary exposition and clichés; their detect-then-edit pipeline improves model text but stays below expert edits in expert rankings (Chakrabarty et al., 2025). Our humanizer and Edit conditions are edits of this kind; the skill differs in applying its audit while writing and in forbidding invented facts, and we evaluate it with model judges rather than experts. A study of "slop" built on an expert taxonomy finds that syntactic features predict slop judgments better than lexical ones, though weakly (AUC 0.64 against 0.49) (Shaib et al., 2026). Banning such patterns token by token makes output unusable at about 2,000 patterns, whereas backtracking at inference time or targeted fine-tuning suppresses thousands of them while keeping quality (Paech et al., 2026). AI generation leaves a stable stylometric footprint that AI editing of human text does not reproduce (Shan et al., 2026).

Detecting AI text. Detectors range from trained classifiers (Adam et al., 2026) to zero-shot statistics (Mitchell et al., 2023; Bao et al., 2024; Hans et al., 2024). The RAID benchmark shows that many are nearly inoperable at low false-positive rates and fail under attacks or unseen generators (Dugan et al., 2024). Paraphrasing defeats watermark-, classifier- and zero-shot detectors (Krishna et al., 2023; Sadasivan et al., 2025), and simple evasion, from adversarial prompting to manual edits and machine paraphrasing, lowers already modest accuracy (Perkins et al., 2024; Weber-Wulff et al., 2023). Detectors misclassify non-native English writing as AI at high rates (Liang et al., 2023) and one RoBERTa detector flagged formal human writing at a 76.3% false-positive rate (Creanga and Dinu, 2026); light, guideline-compliant AI editing is flagged more often than humanizer output (Karr et al., 2026). Untrained humans detect AI text at about chance (Clark et al., 2021; Jakesch et al., 2023), while frequent LLM users do far better (Russell et al., 2025). Text watermarks offer a provenance signal that does not depend on style (Kirchenbauer et al., 2023; Dathathri et al., 2024).

LLMs and admissions essays. LLM-written admissions essays are linguistically distinct from those of 30,000 applicants and resemble the essays of privileged applicants (Lee et al., 2025; Alvero et al., 2024); essay content and style already tracked household income before LLMs (Alvero et al., 2021). Estimated LLM use in essays rose sharply in 2024 (Lee et al., 2026), and in one graduate program, essays flagged as AI-written were associated with lower admission (Isley et al., 2026). AI writing help homogenizes writing across authors (Padmakumar and He, 2024; Doshi and Hauser, 2024; Agarwal et al., 2025) and can shift meaning even when limited to grammar (Abdulhai et al., 2026), and heavy rewriting erases authorship signals (Malik and Awan, 2026).

LLMs as judges. Strong LLM judges agree with human preferences about as often as humans agree with each other, but show position, verbosity and self-enhancement biases (Zheng et al., 2023; Wang et al., 2024), and models can recognize and prefer their own outputs (Panickssery et al., 2024). We discuss how these biases bear on our design in the Limitations section.

3 The skill

The released skill (version 8) is a Markdown instruction file (175 lines) with four reference files: a catalog of tells with fixes, a personal-essay guide, academic-register rules, and before/after examples. An agent loads it when a writing task matches its description. We built it from two reviews: a survey of 16 open-source writing-skill repositories (11 of them humanizer or anti-slop skills), which supplied the procedure of Chen (2026), the "never inject" guardrails of Bronsdon (2026), and the minimum-edit principle and portability test of Yang (2026) (all three MIT-licensed), and an evidence-graded catalog of tells assembled from about 40 sources. Version 8 revised the personal-essay guidance, and three lines of the main file, against 24 real admitted essays (Section 5.4).

Audit order. The catalog showed that several well-measured signals are structural (participial tails (Reinhart et al., 2025), corrective framing (measured in Mandarin by Peng et al., 2026), importance flags, uniform sentence length), that syntactic features predict expert slop judgments better than lexical ones (Shaib et al., 2026), and that expert readers, although they cite vocabulary most often, also rely on sentence structure and originality (Russell et al., 2025). The skill therefore audits a draft in a fixed order: content (the "portability test": could this sentence move unchanged into someone else's text?), staging, inflation, stated lessons, rhythm, paragraph endings, tone, vocabulary, and format. A strong tell justifies an edit on one sighting; weak tells, such as a single transition word, only when several cluster.

Hard rules. The skill never adds a fact, name, number, quote, feeling or memory that the user or the source did not provide; where a sentence needs one, it asks or leaves a bracketed question. It keeps every claim and hedge when editing, uses no evasion tricks (typos, rare synonyms, look-alike characters), and makes minimal edits to human-written text.

Over-correction checks. Humanizer rewrites create tells of their own: a blind test by a contributor, linked from the humanizer's README, counted more uniform-rhythm tells (60 to 101) and more aphoristic closers (38 to 61) after the rewrite pass (Lenk, 2026). The final check lists the last sentence of every paragraph and rewrites when more than two are short or quotable, and checks that a personal essay still has an arc and a line of reflection.

Modes. *Write* runs the full procedure; for personal essays it drafts in two passes, first for shape and then as an editor of its own draft (Section C). *Edit* applies the audit to the user's own draft with minimal changes. *Rebuild*, for drafts the user says an AI wrote, first checks every fact, quote and reading against the user's material; if the draft was built from that material it keeps the checked evidence and writes fresh sentences, and if it was built from the topic alone it treats the draft's examples and plan as the stock answer and writes from the first step. *Detect* quotes passages and names patterns without guessing authorship. *Coach*, used for application essays, interviews the student, proposes a structure, critiques drafts and lists corrections, and does not write the essay or example sentences built from the student's story.

4 Evaluation

We evaluated the skill in three rounds, after a seven-brief pilot that shaped versions 1–4 (Table 4): a development round on 20 briefs, a version comparison on the same briefs, and a held-out round on 16 new briefs written after versions 4 and 6 were frozen. Version 7 was assembled after the held-out round. A second benchmark, built from real admitted essays, measures how human model judges find the texts; version 8 was developed on it and tested only on admissions essays. A separate experiment tests Coach mode.

Briefs. The development set has 20 briefs in five genres, four each (Appendix A, Table 3). Four fictional students, each with about 1,100 to 1,600 words of interview notes and answers to the skill's depth questions, supply the material for a Common App personal statement and a 250-word supplemental answer. The other briefs are 11th-grade argument essays and science-club explainers with no

facts file, and literary analyses restricted to four passages from a public-domain novel or play, copied from Project Gutenberg and checked by script. The held-out set has 16 briefs: two new students (a personal statement and a short answer each), two arguments, two explainers, two literary analyses, and six briefs in five genres outside the development set: two emails (an extension request to a teacher and a cold email to a professor), a cover letter, a team blog post, the discussion section of a lab report, and a school news article. The seven-brief pilot that shaped versions 1–4 had included a first-person blog post and a research-report methods section. Every brief names the only permitted source of specifics; the people in the facts files are fictional.

Conditions. Fresh [generator model, name and version] agents wrote every text. **Plain** (P): the model's default writing. **Word list** (L): the same request plus an instruction to avoid 24 common AI words, six stock phrases and constructions, and em dashes and to vary sentence length. **Humanizer edit** (H): the most-starred of the 11 humanizer skills we surveyed (about 59k GitHub stars on 20 October 2026) (Chen, 2026), applied to P's draft. It edits a draft rather than writing from the brief, so its like-for-like comparison is with the skill's Edit or Rebuild mode on the same draft, not with Write mode. The skill conditions were the Write mode, the Edit mode applied to P's draft, and from version 5 on a Rebuild mode for drafts the user says an AI wrote. Every prompt carried the same brief and facts-file instruction; on briefs about fictional people, the skill and humanizer prompts also said the people were fictional, so that the skill would write the piece instead of switching to Coach mode. The development round compared P, L, H, Edit and Write (version 4); the version comparison compared P, L, version 4 Write, version 5 Write and version 5 Rebuild; the held-out round compared P, L, H, version 4 Write, version 6 Write and version 6 Rebuild.

Judges. Judge agents read all texts for a brief in a shuffled order under neutral labels (T1, T2, . . .), but each judge opened the texts by file path, and the path contained the condition's folder name (a single letter such as P or H; D0, RS, RP, FS or FP in the coaching experiment). Blinding was therefore imperfect: in 6 of the 128 writing-panel judgments the explanation names each text's condition letter (for example "T3 (L)"), and the coaching counselor labelled every piece of feedback by its path. The judges were a strict editor, and the genre's real reader (an admissions officer, an English or science teacher, the recipient of the email, a hiring manager, a team mentor, a newspaper adviser). Each ranked the texts, scored each from 1 to 10 on quality, human-likeness, specificity, voice and structure, rated AI suspicion from 1 to 5, and quoted every claim not supported by the facts file (or every invented statistic, source or factual error when there was none), a coarser, judge-based version of checking generated facts against a knowledge source (Min et al., 2023). In the held-out round a third judge, a second genre reader, ran on a different model of the same family ([second judge model, name and version]). We report mean rank with bootstrap confidence intervals over judgments, a Friedman test across conditions on per-brief mean ranks, and pairwise Wilcoxon signed-rank tests on per-brief mean ranks, so the unit of every test is the brief (16 held-out), not the judgment (48). Judgments of one brief rate the same texts, and the two same-model judges agreed closely (mean Kendall τ 0.78), so the intervals understate uncertainty: resampling briefs instead widens version 4's held-out interval from [1.62, 2.23] to [1.52, 2.38]. Over the 15 held-out pairwise tests, Holm's correction keeps every comparison we report as significant below an adjusted p of 0.025.

Detectors. We scored the development-round and held-out texts with three open-source classifiers: D1, a DeBERTa-v3-large model trained on RAID (Desklib, 2025); D2, a RoBERTa-base model (Fakespot, 2025); and D3, a RoBERTa-large model trained on a RAID subset plus generations that include Anthropic-model output (SuperAnnotate, 2024). Texts are split into chunks of about 350 words, and document scores are length-weighted means. We report the share below 20%, the threshold below which Turnitin shows only an asterisk instead of an AI score (Turnitin, 2026).

Human baselines. To see how the detectors treat human writing, we scored three sets collected before ChatGPT: 16 admitted college essays published by Hamilton College (classes of 2018 and 2022; Hamilton College Office of Admission, 2026) and Connecticut College (class of 2025; Connecticut College Office of Admission, 2026); 18 high-scoring argument essays from the PERSUADE 2.0 corpus (Crossley et al., 2024) (15 with the top score of 6, three with 5); and three

excerpts from two science explainers published in 2020 (one by Bert Hubert, two from one article in The Conversation).

Human-anchored benchmark. Ranking model texts against each other cannot show whether any of them reads like a person, so we built a second benchmark from real writing. It uses 40 essays that Hamilton, Johns Hopkins, Tufts, MIT and Connecticut College published as admitted applicants' work (Hamilton College Office of Admission, 2026; Johns Hopkins University Office of Undergraduate Admissions, 2026; Tufts University Office of Undergraduate Admissions, 2026; MIT Admissions, 2026; Connecticut College Office of Admission, 2026), all from classes of 2026 or earlier and so written before ChatGPT's release. We drew 40 at random from the 90 we collected (28 personal statements of 350 to 700 words, balanced across schools, and 12 supplemental essays) and split them at random, by type, into 24 development and 16 test essays before any judging; four test essays whose judge comments we read early were swapped, at random, with development essays of the same type. For each essay an agent wrote the interview notes a counselor would have taken before the student wrote it: every fact, quote and feeling, regrouped by topic, without the essay's order, images or closing lines, and with no run of more than six words copied outside quoted speech. Each condition's essay was written from these notes alone at the real essay's length: plain (P), word list (L), the skill's version 7, and later versions. A first design, finding the real essay among four essays telling one story, failed because the three model essays shared paraphrases of the same notes (all 20 judgments found it). We switched to judging each essay on its own. Each judge read piles of four essays by four different students, was told that any mix of real and model essays was possible, and gave each essay a probability that a real student wrote it, $p(\text{real})$ from 0 to 100, a quality score from 1 to 10, and the evidence. No pile held two essays about one student, and every text was normalized (straight quotes, single spaces, no title). Each of three judges rated every text, in its own random piles: an editor and an admissions reader on [generator model, name and version] and a second admissions reader on [second judge model, name and version]. We report the share judged real ($p(\text{real}) \geq 50$), the area under the ROC curve for telling the real essays from each version (AUC; 0.5 is chance), and differences paired by essay with 95% bootstrap intervals over essays. A blind checker, given the notes and each student's model essays under neutral labels, quoted every claim the notes do not support. Coder agents, blind to version, marked which of 16 cues each judge rationale cited and in which direction. Three candidates were built on the development essays in three rounds; the second, released as version 8, was frozen, and its content hash recorded, before any test essay was written with it.

Coaching experiment. For each of the six fictional students, an agent playing the student wrote a first draft of the personal statement from their notes, with no style guidance. The draft then received feedback from the skill's Coach mode (version 6 files) or from the plain model asked for feedback, and the student agent revised it once with each set of feedback. Two blind judges ranked the first draft and both revisions, and a counselor agent rated both pieces of feedback for usefulness and counted the essay-ready sentences each wrote for the student. We also measured how much of each revision was copied from the feedback it received: the share of the revision's word 6-grams that occur in the feedback but in neither the first draft nor the student's notes.

5 Results

5.1 Held-out briefs

Table 1 and Figure 1 give the main result. The 16 held-out briefs, in ten genres, were written after versions 4 and 6 were frozen, but we chose which of the two to release after seeing these results, so version 4's figures are the better of two candidates and somewhat optimistic for the versions built on it (7 and 8); version 6, the other candidate, also beat all three baselines (Table 1). Version 4's Write mode ranked first in 23 of 48 judgments and had a mean rank of 1.92 out of 6, against 5.65 for plain generation (Friedman $\chi^2 = 51.0$, $p < 10^{-8}$). It ranked above plain generation in 47 of 48 judgments, above the word-list prompt in 43 and above the humanizer edit in 44 (Wilcoxon $p \leq 0.003$ for each). It made 1.17 unsupported claims per text, against 3.38 for plain generation and 2.98 for the word-list prompt. The Rebuild mode, applied to the plain draft, ranked above the humanizer applied to the same draft in 42 of 48 judgments ($p = 0.0008$) and had the fewest unsupported claims of any condition (0.79).

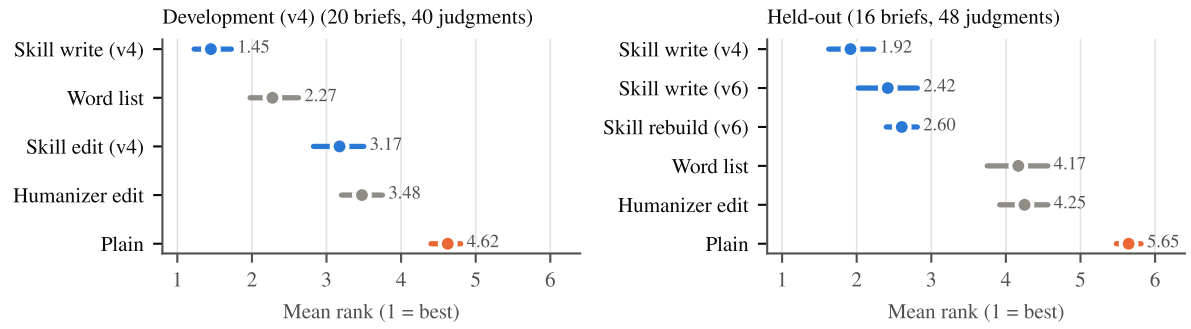


Figure 1: Mean rank with 95% bootstrap intervals (1 = best). Left: development briefs, five texts per panel. Right: held-out briefs, six texts per panel, three judges each. Blue: the skill; orange: plain generation; grey: other approaches.

Condition	Rank	1st	Human	Unsup.
Plain	5.65	0	4.48	3.38
Word list	4.17	3	6.48	2.98
Humanizer edit	4.25	2	5.69	1.88
Skill Write (v4)	**1.92**	**23**	**7.15**	1.17
Skill Write (v6)	2.42	17	6.96	0.96
Skill Rebuild (v6)	2.60	3	7.10	**0.79**

Table 1: Held-out evaluation: 16 briefs, three judges each (48 judgments). Rank: mean rank among six texts (1 = best); 1st: first places; Human: judged human-likeness (1–10); Unsup.: claims per text that judges found unsupported by the facts file or wrong.

Version 6, which added rules derived from a diagnosis of the development judgments, did not improve on version 4: it ranked above version 4 in 21 of 48 judgments ($p = 0.35$). Its argument essays were built around a worked numerical case; one judge called its Write-mode essay on graded homework "dry and mechanical," and another called its Rebuild of the same brief "arithmetic-heavy for an English class." We therefore built version 7 (since superseded by version 8): version 4's Write procedure, with six examples that came from development briefs replaced, plus version 6's Rebuild mode, adapted so that drafts built from the topic alone go to version 4's Write procedure (Section 3). Version 7 as a whole did not go through the held-out panels; its only direct check is a smoke test on three held-out plain drafts, so the held-out numbers describe its parts.

The third judge, on a different model, gave the same order (version 4 1.94, plain 5.38).

By genre (Appendix B), the word-list prompt beat it on the blog post and, narrowly, the cover letter.

5.2 Development round and the version comparison

On the 20 development briefs, version 4's Write mode ranked first in 28 of 40 judgments (mean rank 1.45 of 5; plain 4.62; word list 2.27; humanizer 3.48; Edit mode 3.17). Version 5 (a plain student voice for school essays, a Rebuild mode, a personal-essay checklist) raised judged human-likeness (7.45 against 6.90) but ranked below version 4 in 28 of 40 judgments ($p = 0.055$): judges caught simplified reasoning, and the rule's example stance line became the opening of three of four argument essays. The comparison also exposed a panel effect. Identical plain and version 4 personal statements ranked 8–0 for version 4 in one panel and 3–5 in another; the first panel also held two edits of the plain draft, and judges may have marked the whole family down. The held-out panels again include an edit of the plain draft (44% shared word 6-grams), so version 4's held-out lead on personal statements and short answers (12 of 12 judgments) may be inflated the same way.

5.3 Detectors and human writing

Figure 2 (Appendix E) shows the held-out scores. All 16 texts written with version 4's Write mode and all 16 version 6 rebuilds scored below 20% on all three detectors (version 6's Write mode: 15 of 16), against 9 of 16 for plain generation and 12 each for the word-list prompt and the humanizer edit. The development round agreed (17 of 20 Write-mode texts against 9 of 20 plain; the Edit mode passed 10 of 20). The advantage came from the school and technical genres: on personal statements and short answers every condition passed, plain generation included, with no score above 3% on the held-out briefs and none above 6.2% in the

development round. The human baselines went the other way. None of the 16 admitted college essays scored below 20% on D3, whose mean for them was 61%, and only 5 scored below 20% on both D1 and D2; 2 of the 18 PERSUADE essays passed all three. On personal essays written from our invented briefs these detectors passed every model-written text (89 of 89 across all rounds) and flagged most of the human ones. That pass rate did not hold for real material: written from notes on real admitted essays, only 4 of 40 plain essays and 21 of 40 version 7 essays passed (Section 5.4). This extends earlier reports of false positives on non-native and formal human writing (Liang et al., 2023; Creanga and Dinu, 2026) to admissions essays, on a small sample that the colleges selected and edited, and the human and model essays differ in source material as well as in authorship.

5.4 Against real essays

Development. On the 24 development essays, version 7's essays differed from the real ones in ways its own rules amplified (the notes are speech, and plain generation also used "really" 13.7 times per 10,000 words). They used "really" 21.0 times per 10,000 words (real essays 4.7) and "kind of" 12.1 times (real essays never), twice the contractions, shorter sentences (15.5 against 17.9 words), and stated a lesson 0.2 times per essay against 1.2. Told to sound like "a smart teenager talking" and not to write "taught me", the model carried the notes' spoken register into the essay and ended on a shrug ("I don't actually know.") where admitted students ended on a conviction. In the first round (360 ratings), every real essay was judged real in every rating, the two [generator model, name and version] judges put every real essay above every model essay (AUC 1.00), version 7 was not clearly more human than the word-list prompt (+2.0 [−2.9, +7.1]), and the real essays got the lowest quality scores (5.97 against 6.14 to 6.42). In the coded rationales, tidy structure (a neat arc, motifs and callbacks, everything tied together) was cited as a sign of a model in 75% to 86% of ratings of every model version and in 6% of ratings of real essays. A stated moral was cited as a sign of a real student in 54% of ratings of real essays and as a sign of a model in about half of ratings of model essays, so judges may read a moral through their overall impression of the essay, or through its texture.

	Real	p	AUC	Qual.	Unsup.	Det.
Admitted student	100%	86.4	–	6.21	–	1
Plain	38%	44.8	0.96	6.38	3.06	2
Word list	38%	46.7	0.96	6.35	2.69	5
Skill v7	46%	51.9	0.91	6.29	0.94	10
Skill v8	65%	59.4	0.90	5.98	0.19	6

Table 2: Held-out test: 16 admitted essays never studied during development, 240 single-text ratings by three judges. Real: share of ratings with $p(\text{real}) \geq 50$; p: mean $p(\text{real})$; AUC: separability from the real essays, pooled over judges (0.5 is chance); Qual.: quality 1–10; Unsup.: unsupported claims per essay (blind check against the notes); Det.: essays below 20% on all three detectors, of 16.

Errors are not the gap. The real essays have errors, and judges cited them for 85% of real-essay ratings. A copyeditor agent fixed only outright errors (0 to 6 per essay, under 1% of words). The copyedited essays were still judged real in every rating, $p(\text{real})$ fell by 1.9 points [−3.1, −0.9], and judges could barely tell them from the originals (AUC 0.61). What separates real essays from model ones, then, is mostly not mistakes that a writing aid would have to fake.

Iteration. The development rounds produced three candidates. The first (write, don't talk; state the conclusion) removed the filler but turned stiff, and its round-one gain did not replicate. The second, released as version 8, kept those changes, loosened the structure, stopped adding links between scenes that the notes do not make, and required one whole-hearted conclusion; in round two it beat version 7 by 6.3 points of $p(\text{real})$ [+2.2, +10.5] and cut the share of ratings citing tidy structure from 86% to 51%. The third (8c) added a trying-hard voice, scenes and attempted images: human-likeness tied with version 8 (−1.7 [−5.1, +1.7]) and quality rose 0.18, but it invented about five times as many small details (0.58 against 0.12 unsupported claims per essay; version 7 had 1.50), so we froze version 8.

Held-out test. Table 2 gives the result on the 16 test essays. Paired by essay, version 8 beat version 7 by 7.5 points of $p(\text{real})$ [+2.4, +13.1], the word-list prompt by 12.7 [+6.8, +18.3] and plain generation by 14.7 [+9.3, +20.3]. The two [generator model, name and version] judges judged 47% of version 8's essays real against 22% of version 7's and 9% of plain and word-list essays (version 8 against version 7: +11.4 [+4.5, +18.7]); the third judge called 94% to 100% of every version's es-

says real. Version 8 had no unsupported claim in 13 of 16 essays. Against version 7, quality fell 0.31 points [−0.58, −0.04], and fewer essays passed all three detectors (6 of 16 against 10), plausibly because version 8 moved toward the register of the real essays, which the detectors flag most. The gap did not close. Every rating still judged every real essay real, and the [generator model, name and version] judges separated real essays from version 8 almost perfectly (AUC 0.98).

5.5 Coaching

Both kinds of feedback improved the students' first drafts, and the revision made with Coach-mode feedback ranked above the one made with plain feedback in 7 of 12 judgments (not significant), with fewer unsupported claims (1.67 against 2.75). The counselor judge preferred Coach mode's feedback for all six students (usefulness 8.67 against 7.17 of 10) and found less invented content in it (1.17 items per student against 5.17); plain feedback once supplied what the counselor called "the essay's interpretive center," a line the student never said. Appendix D gives the details and caveats: one counselor judgment per student, longer Coach-mode responses, and version 6's reference files.

6 Discussion

Preferred is not human. The two benchmarks disagree about the skill, though they test different versions and genres. Judges comparing model texts ranked it above plain generation in 47 of 48 judgments; judges reading one essay at a time, with real essays in the pile, found version 7 only slightly more human than a word list (+2.0 points in development, +5.2 on test, both intervals including zero). An anti-AI rulebook defines good writing by what models do wrong, and a model that follows it can land somewhere new: casual, hedged and ending on a shrug, while keeping the tidy structure every model version shared. Some rules had also aged: our generator already avoided "journey" and "challenge", which the admitted essays used more. Version 8 relaxed the rules that pushed essays toward a spoken, hedged register and added a check for engineered structure; relaxing alone (the first candidate) did not replicate, and we did not ablate the rest. The remaining gap is not errors (the copy-edit control) or content (the model essays carried the students' own details); judges point to voice and form: words a notch above speech, uneven passages and digressions, and structure that takes risks. Version 8c's scenes and images did not narrow it and invented more details.

Why content first. The texts judges ranked first argued from one specific idea, used the writer's own facts and ended on content; the last-ranked ones had the genre's stock plan. The word-list prompt changed vocabulary and voice but kept that plan and invented almost as much (2.98 unsupported claims per text against 3.38). We did not ablate the audit order, the no-invention rule or two-pass drafting, and the skill's instructions are far longer than the word-list prompt, so the evaluation shows that the skill as a whole helps, not which part does.

Iterating on judges. Rules distilled from judges comparing model texts did not transfer (version 5's student voice weakened reasoning; version 6's argument rules read as dry and arithmetic-heavy), and each regression was caught only by a same-panel comparison with version 4. Version 8's rules, distilled from judges reading real essays alongside model ones, held on held-out essays (+7.5 points against +6.3 in development).

Detectors in admissions. On personal essays the three detectors got the direction wrong: at least one of them scored 38 of 40 real admitted essays at 20% or more, and on average all three scored the real essays above every model version. One plausible cause is domain: two were trained on RAID, which has no personal essays (Dugan et al., 2024). A score from these detectors is not evidence that an applicant used AI, and since detector scores rise with formal writing (Creanga and Dinu, 2026; Karr et al., 2026), they could fall hardest on applicants who write in the register essay coaching teaches.

7 Conclusion

A style skill that audits content before vocabulary and forbids adding facts won model judges' preference over three baselines, yet its version 7 read only slightly more human than a word list against real admitted essays. Rebuilt on real essays, version 8 raised the share of ratings calling held-out essays real from 46% to 65% and cut unsupported claims to 0.19 per essay, at a cost of 0.31 quality points; real essays stayed separable. Style guidance should be measured against real writing; our survey asks whether human readers agree.

Limitations

Model judges. All judges are language models from the same family as the generator, and models can recognize and favor their own output (Panickssery et al., 2024). Self-preference does not by itself favor one of our conditions, since every condition was produced by the same model, but judges may share the generator's taste for a particular kind of polish. We replaced condition names with neutral labels (though file paths still showed them; Section 4), shuffled the order of texts, used two judge roles (editor and genre reader), and in the held-out round added a judge running on another model of the same family; we did not test judges from other model families or recruit enough human raters. The circularity goes beyond self-preference: agents from the same family drafted the skill, versions 5 and 6 were built from these judges' comments, and the editor prompt in every round, like most reader prompts, cast the judge as someone watching for AI writing ("knows the research on AI-writing tells", "wary of AI-written work"), which is what the skill was built to remove. The held-out round is held out over briefs, not over judges, so our results show what judges from this family prefer. The two same-model judges also agreed with each other (mean Kendall τ 0.78) more than with the third judge (0.55–0.62). A human rating study with the released survey is the most important missing piece.

Human-anchored benchmark. The 40 essays, drawn from 90 that colleges chose to publish, often after editing, are not a representative sample of applicants' writing, and as public web pages they may be in the models' training data, which could help judges recognize them. The notes behind each model essay were reverse-engineered by a model from the finished essay; real notes would be messier, and these notes' spoken register shaped what every condition wrote. All judges were models told that any essay might be AI-written, and the development rounds reused the same 24 essays three times, so the development gains are in-sample; the test result rests on 16 essays. The copyeditor fixed only outright errors, and judges still cited residual slips (a word used slightly wrong, loose logic) in many ratings of the copyedited essays. Judged quality comes from the same model judges and may reward the polish that marks model text.

Test material. The student personas are fictional and were written by a model; real interview notes are likely to be shorter, messier and less quotable. Coach mode was checked on one real student draft and one pressure test, which shows it can work but not how often it does.

Detectors. We used three open-source classifiers. Commercial detectors used by schools (Turnitin, GPTZero) were not tested, so the pass rates here do not predict their scores. The human baselines are small (16 and 40 admitted essays in two overlapping sets, 18 PERSUADE essays) and the admitted essays were selected and edited for publication by the colleges.

Scope and development. One generator model wrote every text, in English, for genres common in US high schools. Versions 1–4 were developed on a seven-brief pilot (Table 4), and five of those briefs (one student's two essays, the phone-ban argument, the mRNA explainer and the Gatsby analysis) are in the 20-brief development set; version 4 also contained six example phrases taken from development briefs. Version 4 ranked about the same on those five briefs (mean 1.40) as on the other fifteen (1.47). Version 5 was built from the development-round judgments and version 6 from both development panels, all on the same 20 briefs, so version 5's version-comparison scores are in-sample. The held-out round uses 16 new briefs, but they were written by the same team that built the skill. Per-genre results rest on one to four briefs each (one brief in each of four held-out genres) and should be read as descriptive; each condition produced one text per brief, so a result on a brief also reflects the luck of that single sample.

Ethical considerations

Dual use. The skill lowers detector scores as a side effect of better writing, and a student could use it to pass off model-written work as their own. We took three steps. The skill uses no evasion tricks (typos, homoglyphs, invisible characters) of the kind studied in the attack literature (Krishna et al., 2023; Sadasivan et al., 2025), but it lowers detector scores anyway: on the six held-out argument, explainer and literary-analysis briefs, every plain text was flagged by at least one detector and every text version 4 wrote passed all three. Its integrity rule sends work a student must submit as their own to Coach mode only when the user says so; our evalu-

ation prompts asked for Write mode on 11th-grade assignments and got finished essays, and a student could do the same with homework. For application essays it coaches and does not write, and it tells the student that the Common App, Yale and many programs count substantive AI-written content as not the applicant's own work. And it tells users that recent model output may carry a provenance watermark (Anthropic, 2026; BleepingComputer, 2026) and that the skill is not a way to hide AI use.

Detector developers. Because we report low detector scores for model text and high scores for human essays, we [notified the developers of all three detectors on <date>], at least 30 days before any preprint.

Human writing and raters. We scored published admitted essays and PERSUADE essays and report only their scores; we do not redistribute their texts, and the released score files identify each essay by source and number rather than by its writer's name, so no named student is linked to an AI score. No real applicant's essay appears in the released data. Any human rating study uses volunteers who give informed consent, collects no names or contact details, and [had approval from <school or teacher>]. Raters under 18 take part only with a parent's or guardian's permission, which the survey asks for but cannot verify. The generated texts from our invented briefs and the survey's answer key are in our public repository (texts derived from real essays are not), so a rater who searched for a passage could learn its condition; the survey asks raters not to.

AI use statement

AI agents ([generator model, name and version], Anthropic) did a large share of this work: they surveyed the literature and existing skills, drafted every version of the skill, generated all evaluation texts, acted as judges, wrote the analysis scripts, and drafted this paper. The development log, prompts, scripts and every intermediate version are released. [The author describes here, accurately, what they did themselves: e.g. which decisions they made, which numbers and citations they checked against the data and sources, and which sections they rewrote.]

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A Briefs

B Results by genre

By genre, version 4 ranked best on personal statements (1.33), arguments (1.33), literary analysis (1.33), the emails (1.67) and the news article (1.00). It ranked worst among the skill conditions on the blog post (4.67), where the word-list prompt won (1.67), and the word-list prompt also edged it on the cover letter (2.00 against 2.33). With one or two briefs per genre these are descriptive. Both losses are in first-person genres, where the audit may strip the looseness the genre wants, but each rests on one brief and three judgments, and the cover-letter gap (2.00 against 2.33) is small. The no-invention rule also did little where there was no source file: version 4's argument essays had 3.17 unsupported or wrong claims per text against 4.00 for plain generation, and its explainers 1.50 against 1.33, so most of the overall drop comes from the genres with a facts or quotes file.

C Personal statements: arc against fidelity

Personal statements were the hardest genre throughout development (Table 4). Plain generation built a better arc by writing connective feelings and causes the student never stated (3 to 11 unsupported claims per essay), and judges preferred it. Forbidding invention flattened the skill's first three versions; one judge wrote that a version 3 essay "mostly walks through the notes in order" and that its climax "lands in paragraph 7 of 9." Two-pass drafting in version 4 (write for shape first, then audit the draft as an editor) recovered part of the arc: the Write mode's first personal-statement win against plain generation on the original notes (1 of 2 judgments), with 4 unsupported claims against 14. Giving both conditions the student's answers to depth questions

Set	Genre (n)	Topics
Development	Personal statement (4)	Four fictional students (Portuguese-American shoe-repair family; Nebraska church organist; Kazakh wild-apple collector; a Houston lion-dance performer), Common App prompts
	Short answer (4)	Same students, 250-word supplements (curiosity, roommate note, activity)
	Argument (4)	Phone bans; test retakes; group-project grades; part-time jobs
	Explainer (4)	mRNA vaccines; seasons; photosynthesis; sky colour and sunsets
	Literary analysis (4)	The Great Gatsby; Frankenstein; Macbeth; The Scarlet Letter (four verified quotes each)
Held-out	Personal statement (2)	Two new students: a Polish e-scooter repairer; an Arizona tamale-stand and weather-station student
	Short answer (2)	Why engineering; the world you come from
	Argument (2)	Later school start; graded homework
	Explainer (2)	Noise-cancelling headphones; leap years
	Literary analysis (2)	Romeo and Juliet; Pride and Prejudice
	Email (2)	Extension request to a teacher; cold email to a professor
	Cover letter (1)	Junior counselor at a library coding camp
	Blog post (1)	Robotics team at a regional competition
	Lab discussion (1)	Pendulum period against length
	News article (1)	Reusable cafeteria trays

Table 3: Writing briefs. Every brief names the only permitted source of specifics (interview notes, a facts file, four quotes, or none for arguments and explainers). All people are fictional. The held-out briefs were written after version 6 was fixed and share no example text with any version of the skill.

Version	Change	vs. plain	Unsup.
v1	audit while drafting	0–4	–
v2	stricter sample-essay rules	0–4	–
v3	countable checks	0–4	1 vs 16
v4	two-pass drafting	1–3	6 vs 19

Table 4: Personal-statement iterations in development (pairwise judgments of better writing, skill wins–plain wins; depth notes for v3–v4; unsupported claims summed over the same pairs, skill vs plain). Small samples: four judgments (two text pairs, two judges) per row. v1–v2 used the original notes, where unsupported claims were 16 vs 29 and 6 vs 33.

raised everyone's quality and halved plain generation's invention. On the two held-out students, version 4 ranked first in 5 of 6 personal-statement judgments with 0.67 unsupported claims per essay, against 4.83 for plain generation.

D Coaching experiment: details

Both kinds of feedback improved the first drafts: the draft ranked last in all 12 judgments. The revision made with Coach-mode feedback ranked above the one made with plain feedback in 7 of 12 judgments (not significant), with fewer unsupported claims (1.67 against 2.75) and lower AI suspicion (1.83 against 2.08). The clearer difference was in the feedback itself. The counselor judge preferred Coach mode's feedback for all six students (usefulness 8.67 against 7.17 of 10) and found less invented content in it (1.17 items per student against 5.17). Neither kind of feedback wrote many whole sentences for the student (1.33 per response for plain feedback; 0.67 for Coach mode, whose rules forbid any, with three in one response), but plain feedback supplied the essay's interpretation: one response supplied what the counselor called "the essay's interpretive center," a line the student never said, and framed it as his own answer; the counselor counted this as substantive content of the kind admissions policies forbid (Common App, 2023; Yale Undergraduate Admissions, 2026). The simulated students copied almost nothing from either feedback (on average under 1% of word 6-grams; at most 3.7% in one revision), so this experiment cannot measure what a rushed real student would copy. Coach mode's responses were also about 1.5 times longer (2,263 words against 1,468), which may be more than a student will read and may have helped them with the counselor, since model judges tend to favor longer answers (Zheng et al., 2023). The feedback comparison rests on one counselor judgment per student, six in all, from a model that was told the Common App rule Coach mode cites and that could see each condition's folder name in the file path. The experiment ran version 6's personal-essay reference, whose Coach-mode section differs from the released version's, and four of the six students are development personas.

E Detector scores

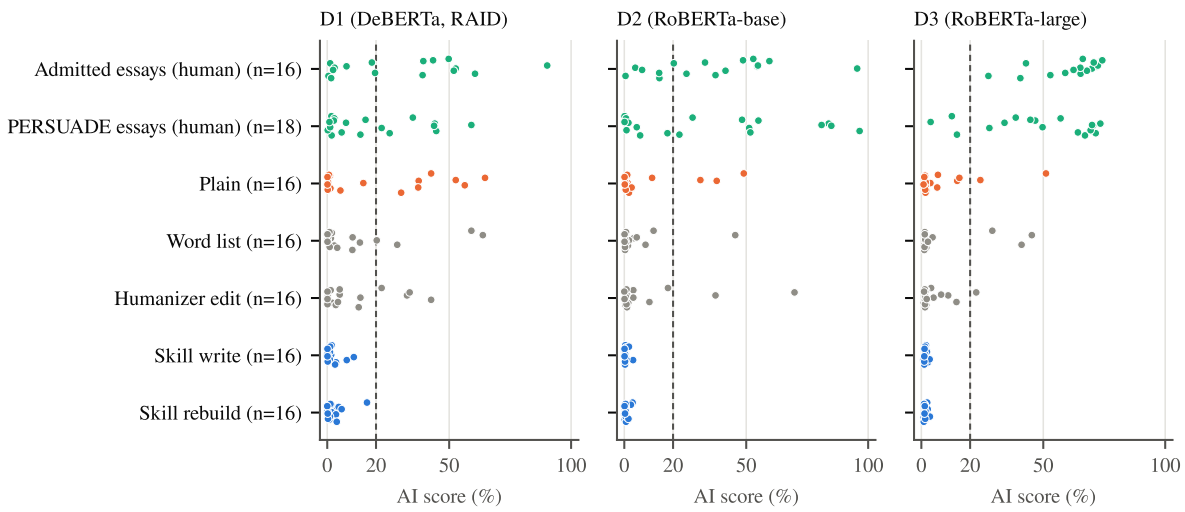


Figure 2: Detector scores per text on the held-out briefs (Skill write: version 4; Skill rebuild: version 6), with human writing collected before ChatGPT for comparison (green). D1–D3 are the detectors described in Section 4. The dashed line is 20%. D3's scores saturate near 73%.